

Dominik Szewczyk

Data Scientist · Machine Learning & AI

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open to relocation



PROFILE

Data Science and AI graduate (cum laude) who builds machine learning systems end to end, from computer vision and NLP models through to dashboards and cloud deployment on Azure with Docker and FastAPI. Across two industry internships and applied research in logistics, plant science, urban planning, and media, I have a habit of turning research into deployed tools and shipping measurable results.

// 01 Experience

Data Scientist, Graduation Internship · KapiteinLabs 2026

- Sole data scientist on a six-month graduation project, taking a predict-and-optimize system from idea to a deployed **no-code Streamlit dashboard**.
- Modelled a plasma reactor's ozone output from its physical settings (geometry, voltage, repetition rate, gas flow, pulse width) using **438 measurements** from TU Eindhoven.
- Built an inverse optimiser that recommends the settings needed to hit a target output, replacing slow one-off lab experiments with instant estimates, and documented it in a graduation thesis.

Data Scientist, Internship · De Rooy Transport & Logistics Feb - Jun 2025

- Built a **multi-model pipeline** forecasting where, when, and how large unexpected pickups will be (over 82% of daily assignments), wired into the planners' PTV route software.
- Pickup occurrence: a Keras model over 5 km grid cells reaching **90.2% accuracy**, 88.9% precision, and 86.5% F1; location via random forest with an **18 m mean error** and 100% of predictions within 1 km; insertion time estimated with XGBoost.
- Engineered spatial, temporal, and holiday features for robustness at peak periods. Estimated impact of up to **€1,500, 1,500 km, and 945 kg CO₂ saved per day**. Featured in the LSB whitepaper.

Web Developer · Freelance Ongoing

- Built and deployed **10 live websites** for small-business clients, owning each project from design through to deployment and hosting.
- Worked directly with clients to scope requirements, iterate on design, and ship the final sites.

// 02 Education

Breda University of Applied Sciences · Data Science & Artificial Intelligence 2022 - 2026

Graduated cum laude. Coursework and projects across machine learning, computer vision, NLP, and MLOps, with two industry internships and a research-based graduation thesis.

II LO im. Andrzeja Frycza Modrzewskiego · Extended Mathematics 2019 - 2023

One of the top-ranked secondary schools in the region, with an extended mathematics profile.

// 03 Skills & Tools

MACHINE LEARNING & AI	Machine Learning, Deep Learning, Computer Vision, NLP & Transformers, Predictive Modeling, Feature Engineering, Reinforcement Learning
LANGUAGES & TOOLS	Python, scikit-learn, TensorFlow, Keras, XGBoost, FastAPI, Streamlit, Power BI, R, Git
CLOUD & MLOPS	Microsoft Azure, Azure ML, Docker, CI/CD pipelines, versioned assets
WAYS OF WORKING	Agile (Scrum), data visualisation, stakeholder communication, translating research into deployed tools
LANGUAGES	English (fluent) · Polish (native) · German (basic)
INTERESTS	Football, gym, chess (3rd category), half marathon (1:43)

// 04 Selected Projects

A selection of applied machine learning, computer vision, and analytics work across industry partners and university research. More detail and live demos at dominikszewczyk224180.github.io.

AI 3D Digital Twin & Object Removal

[Antea Group & Kavel 10](#) · 2024 to 2025

Research on reconstructing urban environments from aerial and drone imagery for city planning. Benchmarked DeepFill v2, LaMa, and PowerPaint for generative object removal and U-Net, DeepLabV3, and FCN for semantic segmentation, then generated textured 3D meshes with Instant NeRF and NeRF Studio and built a VR walkthrough in Unreal Engine.

[Computer Vision](#) · [NeRF](#) · [Semantic Segmentation](#) · [3D Reconstruction](#)

Plant Analysis with Computer Vision & Robotics

[NPEC](#) · 2023 to 2024

Segmented seeds, shoots, and roots from plant images with a U-Net and instance segmentation, localising root tips to within roughly 1 mm. Simulated a PID-controlled liquid-handling robot and applied reinforcement learning to dispense liquid precisely at the detected root tips.

[Computer Vision](#) · [U-Net](#) · [Reinforcement Learning](#)

ATHENA: MLOps for Plant Science

[NPEC](#) · 2024

Took research models to production for ATHENA, a Python package and cloud app for organ segmentation and landmark detection on high-resolution plant images. Built Azure data and training pipelines with versioned assets, containerised with Docker, and served the models via FastAPI. Work recognised publicly by NPEC.

[Microsoft Azure](#) · [Docker](#) · [FastAPI](#) · [MLOps](#)

Emotion Classification with NLP

[Banijay Media](#) · 2024

Detected emotions in video content by classifying text extracted from clip fragments. Compared Naive Bayes, logistic regression, HMM, CRF, RNN, and LSTM, then fine-tuned BERT to reach an F1 above 0.88 on the held-out test set.

[NLP](#) · [Transformers \(BERT\)](#) · [Feature Engineering](#)

Neighbourhood Safety & Crime Prediction

[Municipality of Breda](#) · 2023

A five-person Scrum project for the Municipality of Breda. Built a machine learning model that forecasts monthly crime rates per neighbourhood from historical data, paired with research into the drivers of neighbourhood safety, delivered through an interactive Streamlit dashboard for non-technical stakeholders.

[Machine Learning](#) · [Forecasting](#) · [Streamlit](#)

AI in Facility Management Studies

[BUas](#) · 2023

A mixed-method study into the impact of AI on facility management. Designed a Qualtrics survey, ran in-depth interviews, and analysed the data in R, then led the build of a Streamlit dashboard that was featured at the AI at BUas conference.

[R](#) · [Mixed-method Research](#) · [Streamlit](#)

Renewable Energy Across Europe (SDG 7)

[SDG Hub, BUas](#) · 2022

Analysed renewable energy consumption across Europe for Sustainable Development Goal 7, identifying countries that exceed the global average. Built an interactive Power BI dashboard and surfaced a strong upward trend with a correlation of 0.82 over time.

[Power BI](#) · [EDA](#) · [Data Visualisation](#)

Audience & Content Performance Analysis

[Banijay Media](#) · 2022 to 2023

Analysed content and ratings data to support media strategy. Cleaned and linked the data with a lookup table, then studied audience preferences, top-performing hosts, and the correlation between viewership and social media trends, giving the team a clearer picture of content performance.

[Python](#) · [pandas](#) · [Time Series](#) · [Social Media Analytics](#)